

作業系統作業 HW3

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CH7 Project 1

- 建立一個固定大小的 thread pool
- 有一個 **task queue** 存放待執行的任務
- Worker thread 從 queue 取出任務執行
- 用 **mutex + semaphore** 保護 task queue

```
Thread pool initialized with 4 workers.

[Main] submitted task 0
[Main] submitted task 1
[Worker 0] executing task...
[Task 1] running, sleeping 1s...
[Main] submitted task 2
[Main] submitted task 3
[Worker 1] executing task...
[Task 0] running, sleeping 1s...
[Main] submitted task 4
[Worker 2] executing task...
[Task 3] running, sleeping 1s...
[Main] submitted task 5
[Main] submitted task 6
[Main] submitted task 7
[Worker 3] executing task...
[Task 2] running, sleeping 1s...
[Task 1] done.
[Worker 0] executing task...
[Task 4] running, sleeping 1s...
[Task 0] done.
[Worker 1] executing task...
[Task 5] running, sleeping 1s...
[Task 3] done.
[Worker 2] executing task...
[Task 6] running, sleeping 1s...
[Task 2] done.
[Worker 3] executing task...
[Task 7] running, sleeping 1s...
[Task 4] done.
[Task 5] done.
[Task 6] done.
[Task 7] done.
[Worker 0] shutting down.
[Worker 3] shutting down.
[Worker 1] shutting down.
[Worker 2] shutting down.

Thread pool shutdown complete.
```

ch7_1.png

CH8

5 個 customer thread 持續隨機請求和釋放資源，每次請求前先跑 Safety Algorithm 確認分配後系統仍在 safe state，用 mutex 保護共享的 available、allocation、need 陣列。

```

kelen@host:~/文件/GitHub/OSHW/HW3/OSHW_3_teambase
=== Banker's Algorithm ===
Customers : 5
Resources : 3 types -> 10 5 7

[Customer 0] request granted
[Customer 1] request granted
[Customer 2] request denied: unsafe state
[Customer 3] request denied: unsafe state
[Customer 4] request denied: not enough resources
[Customer 0] released resources
[Customer 3] released resources
[Customer 1] released resources
[Customer 2] released resources
[Customer 4] released resources
[Customer 0] request granted
[Customer 1] request granted
[Customer 3] request granted
[Customer 2] request denied: not enough resources
[Customer 4] request denied: unsafe state
[Customer 0] released resources
[Customer 1] released resources
[Customer 3] released resources
[Customer 2] released resources
[Customer 4] released resources
[Customer 0] request granted
[Customer 1] request granted
[Customer 3] request denied: not enough resources
[Customer 2] request granted
[Customer 4] request denied: not enough resources
[Customer 0] released resources
[Customer 1] released resources
[Customer 3] released resources
[Customer 2] released resources
[Customer 4] released resources
[Customer 0] done.
[Customer 1] done.
[Customer 3] done.
[Customer 2] done.
[Customer 4] done.

=== All customers done ===

```

ch8.png

CH9

- 記憶體大小從 **command line** 傳入
- 支援四個指令：**RQ** (request)、**RL** (release)、**C** (compact)、**STAT** (report)
- 支援三種分配演算法：**F** (first-fit)、**B** (best-fit)、**W** (worst-fit)
- 初始化

- 分配記憶體 (三種演算法各一個)

```
_112590004/CH9$ ./allocator 1048576
=== Contiguous Memory Allocator ===
Total Memory: 1048576 bytes

allocator> STAT

--- Memory Status ---
Addresses [0:1048575] Unused
-----

allocator> RQ P0 100000 F
Allocated 100000 bytes for P0 at address 0
allocator> RQ P1 200000 B
Allocated 200000 bytes for P1 at address 100000
allocator> RQ P2 300000 W
Allocated 300000 bytes for P2 at address 300000
allocator> STAT

--- Memory Status ---
Addresses [0:99999] Process P0
Addresses [100000:299999] Process P1
Addresses [300000:599999] Process P2
Addresses [600000:1048575] Unused
-----

allocator> |
```

CH9.png

- 釋放記憶體
- Compact

- 結束

```
allocator> RL P1
Released memory for P1
allocator> STAT

--- Memory Status ---
Addresses [0:99999] Process P0
Addresses [100000:299999] Unused
Addresses [300000:599999] Process P2
Addresses [600000:1048575] Unused
-----

allocator> C
Memory compacted. Free space consolidated.
allocator> STAT

--- Memory Status ---
Addresses [0:99999] Process P0
Addresses [100000:399999] Process P2
Addresses [400000:1048575] Unused
-----

allocator> X
Exiting.
```